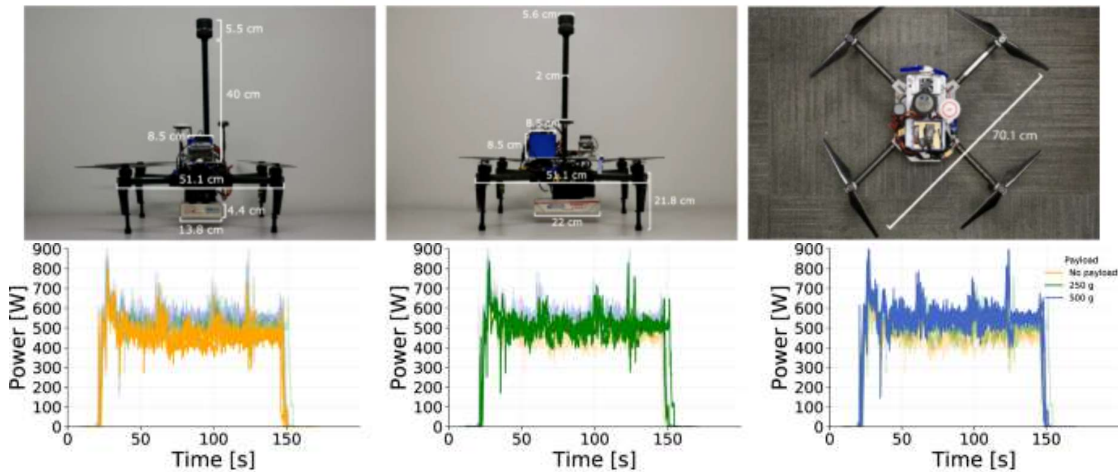


Power Consumption of Robotic Solutions

Case description

Robotic companies must consider power consumption in several regards, e.g., to optimize operating time for battery-powered robots or to minimize the sustainability footprint of robotic solutions. The power consumption of a robot is made up of a complex combination of use by actuators, sensors, computers and communication interfaces. These elements can both be on the robot or be part of supporting infrastructure, e.g., to train AI models and run simulations.

Companies would like to improve operating time as this drives up the business case for robotic solutions. Companies also need to consider the sustainability footprint associated with power use due to increasing requirements for sustainability reporting.



Drone examples for repeated flights with different loads

Dataset example: <https://www.nature.com/articles/s41597-021-00930-x>



Fig. 5. The ECDP was tested using four manipulators, e.g., Franka FR3, Universal Robot UR5e, UR10e and Kinova Gen 3e, placed from left to right. [†] payload weights of these robots are $m_{FR3} = 3$ kg, $m_{UR5e} = 5$ kg, $m_{UR10e} = 10$ kg, and $m_{Gen3} = 4$ kg.

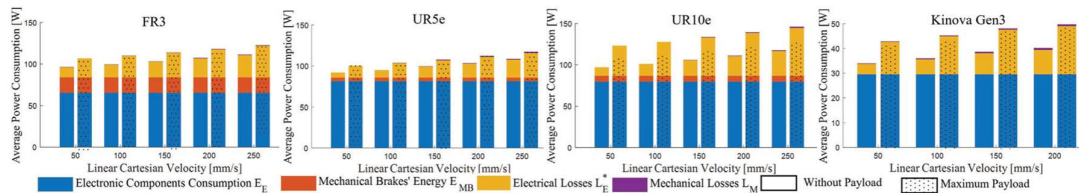


Fig. 6. Average Power Components of each robot of the consumer groups for the four robots.

Cobot examples for different movement speeds

Dataset example: <https://iee-dataport.org/documents/dataset-energy-assessment-collaborative-robots>

Challenge

Develop a software tool to help developers and/or integrators to understand/minimize the power consumption of robotic solutions including their infrastructure.

To get the work started, consider the following:

- Which elements of a robot solution do you consider particularly relevant to optimize?
- Would you focus on battery-powered robots or consider their total sustainability footprint?
- Will your software tool support developers in depth knowledge or provide integrators with overviews?

Keywords

Robot/Drone power optimization, energy efficiency, energy recovery

Tools, methods and materials

The challenge can be addressed with many different methods for data collection, processing and visualization. The data can put different focus on software or hardware elements. Methods can be considered that could alter the programming of the solution or change off-load patterns or perform trade-offs, e.g., accuracy vs. power consumption.

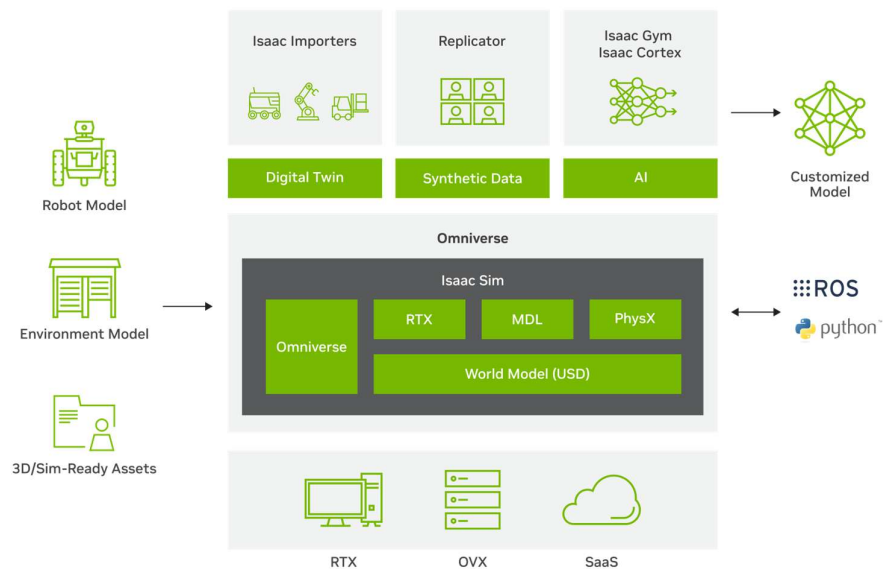
From SDU, the team will receive links to relevant datasets and software frameworks. An edge computing setup with power monitoring is available at the Industry 4.0 lab. In addition, SDU will provide feedback on challenge elements.

Scalable Computing for Robotic Solutions

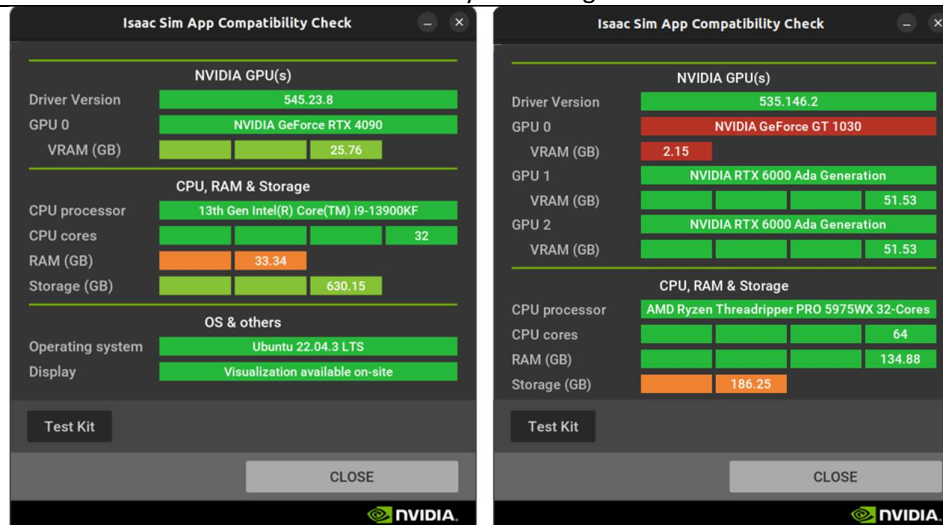
Case description

Further advancement in making robotic solutions more adaptable and intelligent depends on the ability to execute complex simulations for design exploration, synthetic data generation, train large machine learning models and run planning algorithms over many parameters. This results in increasing workloads that robotic companies must cover the cost of. Scalability is the ability of a software system to sustain increasing workloads by making use of additional resources.

To optimize the business case for robotic solutions computing cost has to be minimized to the extent possible. Therefore, efficient scalability is of vital importance for making robotic solutions economic viable.



Nvidia robot controls system design architecture



Nvidia Isacc Sim resource check application

Challenge

Develop a software method or tool to optimize the computing use of robotic workloads in general or a specific workload, e.g. simulation, machine learning, etc.

To get the work started, consider the following:

- What workloads would you like to focus on?
- Would you like to focus on distributed workloads or workloads on a single node distributed over CPU and GPU loads?
- How is it possible to do sensible trade-offs between accuracy vs. cost (e.g. simulation steps, physics accuracy...)
- How do you relate workloads to cost?

Keywords

Distributed computing, SaaS

Tools, methods and materials

The challenge can be addressed with many different methods considering benchmarking, software architecture or distributed computing. The methods can put different focus on software or hardware elements. Methods can be considered that could alter the programming of the solution or change off-load patterns or perform trade-offs, e.g., accuracy vs. cost.

From SDU, the team will receive introduction to relevant software frameworks. Different computing options are available at SDU that can be used for experiments. In addition, SDU will provide feedback on challenge elements.

Developing a Multi-User VR System for Shared Physical Spaces

Case Description

Enterprise VR rooms are transforming physical spaces into shared multi-user VR environments where teams can train together, with companies like Virtualware deploying over 32 VIROO® Rooms worldwide for industries from railways to military training. However, most solutions require expensive custom installations. This project aims to develop an affordable VR system that enables multiple users to share the same physical room while collaborating in a virtual environment. Unlike typical multi-user VR platforms where users connect from different locations, this system addresses the unique challenges of co-located users. Students will identify potential customers in training facilities, educational labs, or team-building venues who could benefit from shared-space VR experiences.

Challenge

The main challenge is creating a stable multi-user VR system where 3 users can operate in the same physical room. While controllers are paired to individual headsets so there's no signal mixing, students must investigate whether Meta Quest 3's inside-out tracking handles multiple headsets in close proximity without interference. The system needs to manage shared physical boundaries, prevent collisions between users who can't see each other, and ensure virtual avatars properly align with users' actual positions. Students must implement basic networking to synchronize the virtual environment while maintaining awareness of real-world space constraints. They will identify which specific use cases benefit from users being physically co-located versus remotely connected.

Keywords

Virtual Reality, Co-located VR, Shared Physical Space, Multi-user Systems, Unity

Tools, methods and materials

The project will involve programming in **C#** and working with **Unity** development. Students will use **Unity's built-in Netcode for GameObjects** networking solution along with VR frameworks like **XR Interaction Toolkit** or **OpenXR**. Development will focus on solving the technical and safety challenges of multiple headsets operating in the same room. Three **Meta Quest 3s** will be provided for testing and development. Code management through **GitHub** with demonstrations throughout the semester.

Developing VR Training Applications with Force Feedback Gloves

Case Description

The haptic technology market is expected to reach a value of \$5 billion by 2028, with force feedback gloves becoming essential for enterprise VR training. Force feedback simulates object hardness, weight, and inertia, enabling trainees to feel the actual weight and resistance of virtual objects. Unlike basic haptic vibrations, force feedback gloves physically restrict finger movement to simulate holding heavy or solid objects. This project aims to develop specialized VR training applications using SenseGlove technology that provides resistance up to 20N of force, equivalent to a 2kg brick on each finger. SenseGlove is already being used by industry leaders including Volkswagen for assembly training, creating opportunities for specialized applications in smaller organizations. Students will identify and engage with potential customers in manufacturing assembly, medical training, or technical education sectors who need realistic weight simulation for effective training.

Challenge

The main challenge is creating training applications where feeling weight and resistance directly improves learning outcomes. Students must integrate SenseGlove hardware with Unity and Meta Quest 2, then develop algorithms that translate virtual object properties (mass, density, material stiffness) into appropriate force resistance. The system must simulate realistic weight distribution when lifting objects, resistance when turning tools like wrenches, or the varying density of different materials. Students will create 1-2 demonstration scenarios (e.g., assembly line training where workers feel component weight, medical training where surgeons feel tissue resistance, or tool operation where users feel proper torque). They will validate which weight-based interactions provide the most training value for their chosen market, targeting organizations that need cost-effective alternatives to enterprise solutions.

Keywords

Virtual Reality, Force Feedback, Weight Simulation, Training Simulation, Unity, SenseGlove, Haptic Resistance.

Tools, methods and materials

The project will involve programming in **C#** and working with **Unity** development. Students will use the **SenseGlove SDK** for Unity integration along with VR frameworks like **XR Interaction Toolkit** or **OpenXR**. Development will focus on creating realistic weight and resistance simulations that enhance muscle memory and procedural learning. One **SenseGlove Nova** and one **Meta Quest 2** headset will be provided for testing and development. Students will identify potential customers and validate market needs through stakeholder analysis. Code management through **GitHub** with demonstrations throughout the semester.

DAO	Dansk: En optimering af FC0 (FC1-FC5) kalkulationen English: An optimization of the FC0 (FC1-FC5) calculation
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Case Description

Dansk: Undersøgelser af dao's Costmodel beregning har vist, at der er tale om komplekse beregninger. I dag tager en beregning mellem 6-8 timer og kræver, at data opdeles i Øst/Vest beregninger. Derudover involverer processen en del manuelt arbejde både før og efter selve beregningen. Formålet med projektet er at undersøge mulighederne for at understøtte løn- og omkostningsberegningen med mere effektive redskaber i fremtiden.

English: An investigation into our Costmodel calculation has made it clear that these are complex calculations. As it works today, a calculation takes between 6-8 hours and is dependent on data being divided into East / West calculations. Prior to the calculations, a lot of manual work is done both before and after the calculation. The purpose is to investigate what options there are to support salary and cost calculations with more efficient tools in the future.

Challenge

Dansk: Udfordringen er at optimere beregningen markant, potentielt via strømlining af data input, begrænset udfaldsrum, eller andre algoritmiske optimeringer. En central del af opgaven er at vælge et bedre og mere effektivt koordinatsystem end det nuværende GPS-baserede WGS84, som kræver meget komplekse beregninger til afstande. Løsningen skal kunne hente data fra forskellige systemer og skal kunne kommunikere med en eksisterende besked-kø. Selvom fuld integration med det interne ERP-system ikke er et krav i opgaven, skal løsningens design tage højde for det. Den endelige løsning skal overholde eksisterende design- og kodestandarder, herunder SOLID-principperne.

English: The challenge is to greatly optimize this calculation, possibly through streamlining data input, limiting the sample space, or other algorithmic technical optimizations. A key part of the task will be to choose a better coordinate system than the current GPS-based WGS84, which is inefficient and requires very complex calculations for determining distances. The solution must be able to collect data from different systems with different interfaces and communicate with an existing message queue. While integration with the internal ERP system is not a requirement in the project, the solution's design must account for it. The developed solution must comply with existing design and coding standards, including SOLID.

Keywords

Dansk: Optimering, Costmodel, Dataintegration, Koordinatsystem, Algoritmer, AI.
English: Optimization, Costmodel, Data Integration, Coordinate System, Algorithms, AI.

Tools, methods and materials

Dansk: De tekniske værktøjer, der skal anvendes, er Java, Spring Boot, RabbitMQ og Kubernetes.

English: The technical tools to be used are Java, Spring Boot, RabbitMQ, and Kubernetes.

DAO	Dansk: Optimering af adressedatabaser English: Optimization of address databases
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Case Description

Dansk: Den nuværende adressedatabase er kompleks, opbygget over lang tid og indeholder mange manuelle kodemæssige operationer, der reducerer fleksibiliteten. Den kan desuden ikke håndtere leverancer til erhvervskunder. Formålet er at undersøge, hvordan man kan opbygge en ny adressedatabase, der understøtter rutedesign og leveringer i et samlet netværk med en optimal datastruktur. Tesen er, at en graf-database kan opnå den ønskede søgningskompleksitet på $O(n^2)$ eller $O(\log n)$. Sekundært indgår beregninger på etageforhold i planlægningen.

English: The current address database is complex, has been built up over a long period of time, and has many manual code-related operations that increase complexity and reduce flexibility. Furthermore, it cannot handle deliveries to business customers. The purpose is to investigate how to build an address database that can support route design and deliveries in a unified network, using a data structure with optimal $O(n^2)$ or $O(\log n)$ complexity for search and neighbor traversal. The thesis is that by using a graph database, these searches can achieve that level of complexity. As a secondary use case, include calculations on floor conditions in the planning model.

Challenge

Dansk: En central opgave er at undersøge og vælge en ny databaseteknologi, da dao ikke ønsker at videreføre den nuværende Oracle-baserede løsning. Den nye løsning skal baseres på adresser fra Danmarks Adresseregister via DAWA API'et. Der skal lægges stort fokus på at optimere læsning, opdatering og berigelse af adresser, da databasen er central for mange systemer. Løsningen skal kunne håndtere forskellige koordinatsystemer (både til GPS-rutevejledning og geokoordinater til afstandsberegning) med fokus på fleksibilitet og optimering. En optionel udfordring er at undersøge brugen af en specialiseret AI-model til at automatisere rettelsen af forkert indtastede adresser.

English: A core task is to research and select a new database technology, as dao does not want to continue with the current Oracle-based solution. The new address solution must be based on the Danish Address Register, using the DAWA API. It is essential to focus on optimizing the reading, updating, and enriching of addresses, as the database is central to many systems. The database must also accommodate the needs of different systems that use various coordinate systems (e.g., GPS for route guidance vs. geocoordinates for distance calculation), focusing on both flexibility and optimization. An optional challenge is to investigate using a specialized AI model to automate the correction of incorrectly entered addresses, a task currently handled by manual resources.

Keywords

Dansk: Adressedatabase, Graf-database, Optimering, Søgealgoritmer, Dataintegration, AI, DAWA.
English: Address Database, Graph Database, Optimization, Search Algorithms, Data Integration, AI, DAWA.

Tools, methods and materials

Dansk: De tekniske værktøjer, der skal anvendes, er Java, Spring Boot, RabbitMQ, GraphDb og Kubernetes.

English: The technical tools to be used are Java, Spring Boot, RabbitMQ, GraphDb, and Kubernetes.

Codex-Agnosco: Local LLM-Assisted Legacy Code Analysis (VS Code Extension)

Case Description

Teams spend significant time understanding legacy code, and many cannot use code assistants due to privacy constraints. This project ships a privacy-first VC Code extension that explains functions, finds related code, and proposes docstrings using a local language model (no code leaves the machine). Students will identify stakeholders such as enterprise development teams in regulated industries and software modernization consultants who need on-prem assistance.

Challenge

Overcome single-file limits by architecting a repository-aware retrieval-augmented pipeline. Parse the repository (e.g., with Roslyn) to extract symbols and structure; chunk meaningful units (methods/classes); compute local embeddings; store them in a lightweight vector index; retrieve top-k relevant code for a selection or question; and prompt a local code-centric LLM to generate explanations, related-code summaries, and docstrings. Provide a VS Code extension UI that calls a local HTTP service; keep performance acceptable on a typical developer laptop.

Keywords

Legacy Code, Static Analysis, Retrieval-Augmented Generation (RAG), Embeddings, Roslyn, VS Code Extension, On-Prem AI

Tools, methods and materials

Extension frontend in Typescript; backend service in C# 12+ (.NET). Use Roslyn for parsing; generate embeddings with a CPU-friendly sentence-transformers model; store in FAISS or SQLite. Run a small, permissively licensed code model locally via Ollama or LM Studio (e.g., StarCoder2 or quantized CodeLlama). Manage on GitHub with iterative demos and tests for indexing/retrieval quality.

Out of scope: multi-language support, model fine-tuning/training, telemetry.

Database Migration Safety Analyzer

Case Description

For any company running a production application, database migrations are high-stakes operations; a single faulty script can cause data loss or downtime. Execution tools (e.g., Flyway) don't analyze safety. This project develops an open-source static analyzer that examines SQL migration scripts before execution to detect dangerous patterns that could lead to data loss, performance degradation, or irreversible changes. Students will identify potential users such as SaaS startups, digital agencies, and SMB DevOps teams.

Challenge

Accurately model the impact of schema changes. Scope to PostgreSQL (MySQL as an optional stretch). Build or extend a DDL parser, transform raw SQL into an AST, and implement a rule engine to detect hazards (e.g., narrowing type changes, adding NOT NULL without a default, dropping populated columns, foreign keys without supporting indexes, non-concurrent index builds, table-rewrite/lock-risk operations). Track schema state across ordered migration files, rank findings by severity/likelihood, and produce an actionable report and CI signal.

Keywords

Database Migration, Static Analysis, SQL Parsing, Schema Evolution, DevOps, PostgreSQL.

Tools, methods and materials

Implement in Python using sqlglot or similar, or in C# 12.0+ (.NET) with an ANTLR Postgres grammar. Analyzer takes a directory of SQL files and emits a risk report (Markdown / HTML) plus a GitHub Actions check that fails on high-risk migrations. Use Dockerized PostgreSQL for optional spot validation. Manage on GitHub with unit tests over a seeded corpus.

Out of scope: cross-dialect support, DML/business-logic analysis, production database access.

Digital Accessibility Compliance Checker

Case Description

Digital accessibility is a legal and ethical imperative. Regulations such as the European Accessibility Act and standards like WCAG create continuous demand for in-house audits that surface high-impact issues and produce clear, shareable reports. This project delivers a lightweight, extensible auditing platform; students will identify stakeholders such as university IT and compliance units, corporate web teams, and disability services that require fast, reliable evidence of barriers and actionable guidance.

Challenge

Design and build a robust, respectful scanning and reporting pipeline. The system must run a curated subset of WCAG checks across a defined set of public URLs while applying privacy-by-design (persist findings only, not page content). Engineer a resilient headless browser integration that honors robots.txt and throttling; inject an accessibility engine; transform raw findings into an intelligent, prioritized report by aggregating violations, ranking by a severity×frequency metric, linking each finding to WCAG guidance, and suppressing obvious false positives to keep results actionable.

Keywords

Web Accessibility, WCAG, EN 301 549, Headless Browser, axe-core, Compliance Reporting, GDPR by Design.

Tools, methods and materials

Implement in C# 12.0+ (.NET) or Python. Use Playwright for headless fetching and inject axe-core for automated checks. Persist results to JSON or SQLite; generate a static HTML report (optional PDF via wkhtmltopdf or WeasyPrint). Students select targets via a small config/CLI (sitemap or hand-picked public pages). Manage code on GitHub with unit/functional tests focused on aggregation/ranking and short iterative demos.

Out of scope: authenticated areas, custom rule authoring, and automatic fixes.